

Cryptocurrency Price Prediction Using Market Indicators

DSA 210 Introduction to Data Science - Final Report

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Project summary. This project studies short-term Bitcoin price movement by combining BTC/USDT hourly OHLCV data with a market-wide USDT indicator. The analysis follows the data science pipeline from data preparation and exploratory analysis to machine learning model evaluation. The final machine learning task predicts whether the next hourly BTC return is positive or negative.

1. Motivation

The motivation of this project is to understand whether short-term cryptocurrency price movements can be partially explained using both price-based variables and broader market indicators. Cryptocurrency markets are highly volatile, so predicting exact future prices is difficult. For this reason, the project focuses on a simpler and more realistic target: predicting whether the next-hour BTC movement is upward or downward. The purpose is not to build a perfect trading system, but to apply data science methods to a real financial dataset and evaluate how much signal can be extracted from noisy market data.

2. Data Source and Collection

The project uses two main data sources. First, BTC/USDT OHLCV data was used as the main price dataset. It includes open, high, low, close, and volume values. The original data was at a high-frequency level and was processed into hourly observations. Second, USDT market indicator data was added to represent market-wide behavior and investor sentiment. The idea is that changes in USDT market share or market cap can reflect movement toward or away from stablecoin positions.

After cleaning and merging, the processed dataset was saved as *final_dataset.csv*. This file contains 45,521 hourly observations and 14 columns, including BTC price variables, volume, USDT market information, returns, volatility, moving averages, dominance change, and volume change.

3. Previously Submitted Components and Reproducibility

The main parts of this project were already included in previous milestone submissions. The proposal described the research idea, data sources, expected enrichment, and planned features. The April 14 milestone included data collection, cleaning, merging, exploratory data analysis, and hypothesis testing. The May 5 milestone included the machine learning notebook, the processed input dataset, and the output files from the ML analysis. This final report brings those already submitted components together in one written summary.

For reproducibility, I included *final_dataset.csv* so the May 5 notebook can be tested and reproduced if needed. The May 5 outputs are saved as *may5_ml_dataset.csv* and *may5_model_comparison.csv*.

4. Data Preparation and Feature Engineering

The data preparation stage consisted of converting the raw cryptocurrency data into a format suitable for analysis. The BTC data was resampled to hourly frequency, then merged with the USDT market indicator data by timestamp. Missing values and invalid infinite values were removed before modeling. This was important because some derived variables, such as percentage changes, can produce infinite values when a denominator is zero.

Several additional variables were created to enrich the dataset. These included return and log return, 24-hour volatility, 24-hour and 72-hour moving averages, USDT dominance or market-cap changes, volume change, lagged return features, technical indicators such as RSI, MACD, and Bollinger Band variables, and time-based features such as hour-of-day and day-of-week transformations. The final ML-ready dataset contains 45,495 rows and 36 columns after cleaning and feature engineering.

5. Data Analysis and Machine Learning Methods

5.1 Exploratory Analysis and Hypothesis Testing

In the April 14 milestone, the project focused on understanding the structure of the combined dataset. The analysis examined BTC price behavior, returns, volatility, volume patterns, and the relationship between BTC movements and the USDT market indicator. Hypothesis testing was used as an initial statistical step to check whether some market conditions were associated with differences in BTC returns or volatility.

5.2 Machine Learning Target

For the May 5 milestone, the prediction task was converted into a binary classification problem. The target variable was defined as 1 when the next-hour BTC return was positive and 0 otherwise. In other words, the model tries to predict whether BTC will move up or down in the next hour. The future return was calculated before creating the target, and the backtesting step used the same future return to avoid mixing the current return with the prediction target.

5.3 Models Used

Three course-level machine learning models were applied:

- Logistic Regression: used as a baseline linear classifier.
- Decision Tree: used as an interpretable nonlinear model.
- Random Forest: used as an ensemble method based on multiple decision trees.

The models were evaluated using a chronological train-test split rather than a random shuffle, because this is time-series market data. This makes the evaluation more realistic because the model is trained on earlier observations and tested on later observations.

6. Findings

The model comparison results are shown below. Accuracy, precision, recall, F1 score, and ROC-AUC were used because the task is binary classification.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.520	0.544	0.371	0.441	0.533
Decision Tree	0.521	0.538	0.446	0.488	0.533
Random Forest	0.525	0.551	0.370	0.443	0.542

Random Forest achieved the highest accuracy and ROC-AUC, while Decision Tree achieved the highest F1 score. Overall, the models performed only slightly above random guessing. This is a meaningful finding rather than a failure, because short-term cryptocurrency price movements are noisy and difficult to predict. The results suggest that the engineered features contain some limited predictive signal, but not enough to produce a very strong short-term classifier.

The feature importance analysis from tree-based models also helped identify which variables contributed more to prediction. In general, lagged returns, volatility-related features, moving averages, volume-related variables, and technical indicators were more informative than raw price values alone.

7. Limitations and Future Work

There are several limitations in this project. First, cryptocurrency markets are extremely noisy, especially at the hourly level. This makes prediction difficult even when using multiple engineered features. Second, the project uses mostly price-based and market-cap-related indicators; it does not include news, social media sentiment, macroeconomic announcements, funding rates, order book data, or liquidation data. Third, the backtesting step is simplified and does not include transaction costs, slippage, exchange fees, or risk management rules. Fourth, the model evaluation is based on historical data, so real-time performance could be different.

For future work, the project could be improved by adding more external indicators, comparing more time intervals, using rolling-window validation, testing more advanced but course-appropriate feature selection methods, and designing a more realistic trading simulation. Another extension would be to compare classification models with regression models that directly predict future return size.

8. Conclusion

This project applied the main steps of a data science pipeline to a real cryptocurrency prediction problem. BTC OHLCV data was enriched with a USDT market indicator, cleaned, merged, and transformed into an ML-ready dataset. Exploratory analysis and hypothesis testing were completed in the earlier milestone, and the May 5 milestone applied machine learning models to predict next-hour BTC direction. The results show that Random Forest performed best in terms of accuracy and ROC-AUC, but all models remained close to random classification. This supports the idea that short-term crypto prediction is difficult, but also shows how data science methods can be used to structure, test, and evaluate a market prediction problem.

9. AI Usage Disclosure

AI tools were used for technical assistance in structuring the notebook, debugging code, checking file organization, and preparing the written summary. The project topic, data choices, analysis decisions,

interpretation of the results, and final review were completed by the student. AI assistance was used as support, not as a replacement for the project work.

10. Data and Project Files

- `crypto_project_proposal1.pdf` - project proposal and initial plan.
- `crypto_analysis_april14.ipynb` - April 14 data collection, EDA, and hypothesis testing milestone.
- `final_dataset.csv` - processed dataset combining BTC OHLCV and USDT market indicator data.
- `crypto_analysis_may5_milestone.ipynb` - May 5 machine learning milestone.
- `may5_ml_dataset.csv` - ML-ready dataset after additional feature engineering.
- `may5_model_comparison.csv` - model comparison output.